

## IMPROD

### Manual for imaging – IMPROD, 3T Verio

0. Ask a patient to go on a toilet to empty his rectum (not bladder - if the bladder is partly filled it is easier to see prostate BUT bladder should not be too big). This is the most important step of all! Inject glucagon/buscopan before examination to suppress bowel movement.
1. Patient positioning is extremely important. Prostate should be between 2 coil elements (between coil element of spine coil and body matrix coil). When patient lay on the table find his anterior superior iliac spine. It is an important landmark of surface anatomy, it refers to the anterior extremity of the iliac crest of the pelvis. Prostate is in the midline about 6-8 cm below anterior superior iliac spine.
2. Register the patient. Use protocol – **IMPROD.edx**
3. Run **localizer\_sag\_tse**
4. Check if prostate is between 2 coil elements. If the rectum is distended due to rectal air ask patient to go on a toilette to empty his rectum. Rectum air causes susceptibility artifacts which significantly decreases image quality of DWI.
5. Start planning **localizer\_tra\_tse**. Important step. Middle of FOV (small yellow circle) should be exactly in the middle of prostate. You can find prostate as a structure under urinary bladder.
6. Apply **localizer\_tra\_tse**. In most of cases table will move a bit. If prostate was exactly in the middle of FOV of **localizer\_tra\_tse** before planning of **localizer\_tra\_tse** the table will not move.
7. Start planning **localizer\_sag\_tse=ref**. Copy table position from **localizer\_tra\_tse**
8. Apply **localizer\_sag\_tse=ref**
9. Start planning **localizer\_cor\_tse=ref**. Copy table position from **localizer\_tra\_tse**
10. After finishing **localizer\_cor\_tse=ref**, delete this: **localizer\_sag\_tse**.
11. Put **localizer\_tra\_tse**, **localizer\_sag\_tse=ref**, **localizer\_cor\_tse=ref** on planning screens.
12. Start planning **t2\_tse\_sag\_320\_p2**. For planning use **localizer\_tra\_tse**, **localizer\_sag\_tse=ref**, **localizer\_cor\_tse=ref**. Middle of FOV (small yellow circle) should be in the middle of prostate. Apply the sequence.
13. Start planning **t2\_tse\_tra\_320\_p2**. For planning use **localizer\_tra\_tse**, **localizer\_sag\_tse=ref**, **localizer\_cor\_tse=ref**. Middle of FOV (small yellow circle) should be in the middle of prostate. Apply the sequence.

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13. Start planning **diff\_tra\_LowSAR\_b0-100-200-350-500**. Copy center of slice position from **t2\_tse\_tra\_320**. Apply the sequence **diff\_tra\_LowSAR\_b0-100-200-350-500**.
14. Start planning **b\_1500**. Copy center of slice position from **diff\_tra\_LowSAR\_b0-100-200-350-500**. Apply the sequence **b\_1500**.
15. Start planning **b\_2000**. Copy center of slice position from **diff\_tra\_LowSAR\_b0-100-200-350-500**.! Apply the sequence **b\_2000**.

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